Leadership and Productivity in Public Schools

Evidence from mandatory principal turnover

Paper idea for discussion. October 2026

Through which inputs does a school manager raise output?

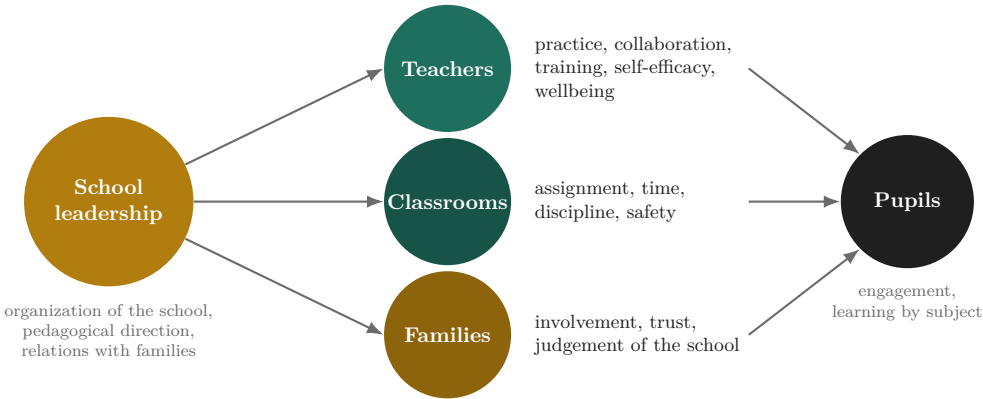
Principal value added on achievement is established. The production process is not: how the managerial input moves teacher effort and practice, classroom organization, and parental inputs.

Literature

Value added	0.05–0.10 SD per SD of principal quality (Branch, Hanushek, and Rivkin, 2012; Böhlmark et al., 2016); 0.03 math, 0.01 reading net of school-year shocks (Bartanen, Husain, and Liebowitz, 2024).
Turnover	−0.03 SD and higher teacher exit after a change of principal (Bartanen, Grissom, and Rogers, 2019).
Mechanism	Selection and retention of teachers (Grissom and Bartanen, 2019; Bartanen and Grissom, 2023; Muñoz and Prem, 2024).
Unidentified	Effects on the practice and effort of a fixed teaching staff, on classroom organization, and on parental inputs.

Recent work

Causal map



Exogenous variation enters at the leadership node. Every other node is observed.

Setting

Managers	Principals are teachers elected for four-year terms, without authority over hiring, dismissal, pay, or curriculum.
Labor input	Fixed across a change of principal. The selection channel is shut; what remains is the managerial input.
Mandatory exits	Retirement at 65 and a two-renewal limit (twelve years), both pre-determined.
Data	Universe of Madrid public primary schools, 2015–2018: principals, teachers, timetables, all pupils and parents by classroom, test scores.

Identification



Instrument	A mandatory exit in 2016, 2017, or 2018, determined by age or tenure.
Exclusion	The exit date affects outcomes only through the change of principal.
Test	No pre-trends in any outcome before the exit.
Benchmark	Voluntary exits, estimated separately: the selection on unobservables the literature cannot rule out.

Reduced form

τ_s: year of the mandatory exit in school s. i: teacher, classroom, pupil, or parent.

$$Y_{ist} = \alpha_s + \lambda_t + \sum_{k \neq -1} \beta_k \, \mathbf{1}[\, t - \tau_s = k \,] + \varepsilon_{ist}$$

$\beta_{k \geq 0}$	Dynamic treatment effect k years after the exit.
$\beta_{k < 0}$	Pre-trends.
Estimator	Staggered adoption; not-yet-treated controls; clustered by school.

Local average treatment effect

$Z_{st} = \mathbf{1}[t \geq \tau_s]$, $D_{st} = \mathbf{1}[\text{new principal}]$.

$$D_{st} = \pi Z_{st} + \alpha_s + \lambda_t + u_{st}$$
$$Y_{ist} = \beta \widehat{D}_{st} + \alpha_s + \lambda_t + \varepsilon_{ist}$$

Compliers	All schools at 65; schools where the tenure limit was binding.
Within unit	α_j for the same teacher under two principals; lagged score for the same pupil, grades 3 to 6.
Mediation	β on achievement, with and without the teacher, classroom, and family outcomes as controls.

Variance of manager effects; heterogeneity

$Y_{st} = \mu_s + \theta_{p(s,t)} + e_{st}$. Around a mandatory exit, θ_{old} is replaced by θ_{new}:

$$\text{Var}(\Delta Y_{st} \mid Z_{st} = 1) - \text{Var}(\Delta Y_{st} \mid Z_{st} = 0) = 2\sigma_\theta^2 \qquad Y_{ist} = \ldots + \beta D_{st} + \delta\, D_{st} \times W_s + \varepsilon_{ist}$$

σ_θ^2	Dispersion of manager effects without ranking managers (Jones and Olken, 2005).
W_s	Pre-determined: successor's practices and wellbeing as a teacher in $\tau_s - 1$; project score; staff rating of the incumbent.
δ	Treatment effect heterogeneity by manager type.

Outcomes and expected timing

Input	Outcome	Source	When
Classrooms	Allocation of teachers and support hours	Timetables	Year 1
Classrooms	Discipline, climate	Pupil and teacher reports	Years 1–2
Families	Involvement, trust, satisfaction	Parent survey	Years 1–2
Teachers	Self-efficacy, wellbeing, collaboration, training	Teacher survey	Years 1–2
Teachers	Classroom practice, within teacher	Teacher survey	Years 2–3
Output	Engagement; achievement by subject	Pupil survey, tests	Last

Pre-registered indices by input.

Open questions

- **Inputs.** Which should respond within the first year of a new principal.
- **Data.** Stable teacher identifier; principal's age and tenure; survey items rating the principal.
- **Sample.** Number of mandatory exits, 2015–2018; appointment records and later years.

References I

Bartanen, B., and J. A. Grissom. 2023. "School Principal Race, Teacher Racial Diversity, and Student Achievement." *Journal of Human Resources* 58(2): 666–712.

Bartanen, B., J. A. Grissom, and L. Rogers. 2019. "The Impacts of Principal Turnover." *Educational Evaluation and Policy Analysis* 41(3): 350–374.

Bartanen, B., A. N. Husain, and D. D. Liebowitz. 2024. "Rethinking Principal Effects on Student Outcomes." *Journal of Public Economics* 234.

Böhlmark, A., E. Grönqvist, and J. Vlachos. 2016. "The Headmaster Ritual: The Importance of Management for School Outcomes." *Scandinavian Journal of Economics* 118(4): 912–940.

Branch, G. F., E. A. Hanushek, and S. G. Rivkin. 2012. "Estimating the Effect of Leaders on Public Sector Productivity: The Case of School Principals." NBER Working Paper 17803.

Fenizia, A. 2022. "Managers and Productivity in the Public Sector." *Econometrica* 90(3): 1063–1084.

Grissom, J. A., and B. Bartanen. 2019. "Strategic Retention: Principal Effectiveness and Teacher Turnover in Multiple-Measure Teacher Evaluation Systems." *American Educational Research Journal* 56(2): 514–555.

References II

Hanushek, E. A., A. Morgan, S. G. Rivkin, J. C. Schiman, A. Shakeel, and L. Sartain. 2024. "The Lasting Impacts of Middle School Principals." NBER Working Paper 32642.

Jones, B. F., and B. A. Olken. 2005. "Do Leaders Matter? National Leadership and Growth Since World War II." *Quarterly Journal of Economics* 120(3): 835–864.

Lavy, V., A. Boiko, and G. Rachkovski. 2023. "Effects and Mechanisms of CEOs Quality in Public Education." *Economic Journal*.

Lemos, R., K. Muralidharan, and D. Scur. 2024. "Personnel Management and School Productivity: Evidence from India." *Economic Journal* 134.

Miller, A. 2013. "Principal Turnover and Student Achievement." *Economics of Education Review* 36: 60–72.

Muñoz, P., and M. Prem. 2024. "Managers' Productivity and Recruitment in the Public Sector: The Case of School Principals." *American Economic Journal: Economic Policy* 16(4): 223–253.

Rahaman, A., and C. Koedel. 2026. "Principal Effects on Teacher Working Conditions." CALDER Working Paper 335.

References III

Tsao, C. 2024. “Managers in Public Schools.” Working paper, Princeton University.

Appendix: recent work on school and public-sector managers

Title	Journal	Year
Principal effects on teacher working conditions (Rahaman and Koedel)	CALDER WP	2026
Rethinking principal effects on student outcomes	J. Public Economics	2024
Managers' productivity and recruitment in the public sector: the case of school principals	AEJ: Economic Policy	2024
Personnel management and school productivity: evidence from India	Economic Journal	2024
The lasting impacts of middle school principals	NBER WP	2024
Managers in public schools (Tsao)	Princeton JMP	2024
Effects and mechanisms of CEOs quality in public education	Economic Journal	2023
School principal race, teacher racial diversity, and student achievement	J. Human Resources	2023
Managers and productivity in the public sector	Econometrica	2022

Tsao: event studies of principal moves and retirements, Kentucky and Texas; effects on scores, absenteeism, teacher retention. Rahaman and Koedel: variance of principal effects on 20 teacher-reported working conditions, Illinois.

Back